University of Central Punjab
Faculty of Information Technology
Midterm lab exam– Spring 2026

Subject	Software Construction and Development	Course Code	SESD3241
Sections			
Allowed Time	90 minutes	% contribution towards final lab evaluation	

INSTRUCTIONS

1. Read the tasks carefully.
2. All questions are mandatory. Only appropriate queries would be entertained by the lab instructor.
3. Exchange of solution is not allowed.
4. Submit the exam solution within prescribed time.

Question 1:

A restaurant billing system wants to calculate the final bill for customers based on multiple food orders.

Requirements:
Take input for number of orders (max 5)
For each order, input food price.

Calculate total bill

Discount Rules:
Total ≥ 8000 → 20% discount
5000–7999 → 10% discount
3000–4999 → 5% discount
Otherwise → No discount

Formula:
Discount = (Total Amount × Discount %) / 100

Output:
Total Bill Amount
Discount Amount
Final Payable Amount

Question 2:

```
public class ParkingFee {
    public void calculateFee(String vehicleType, int hours) {
        if (vehicleType.equals("car")) {
            int fee = hours * 50;
            if (hours > 5) {
                fee += 100;
            }
            System.out.println("Fee: " + fee);
        } else if (vehicleType.equals("bike")) {
            int fee = hours * 20;
            if (hours > 5) {
                fee += 50;
            }
            System.out.println("Fee: " + fee);
        } else {
            System.out.println("Invalid Vehicle");
        }
    }
}
```

Tasks:
a) Identify SOLID Principle
b) Explanation
c) Problem Analysis
d) Design Improvement

Question 3:

```
public class ParkingFee {
    public void calculateFee(String vehicleType, int hours) {
        if (vehicleType.equals("car")) {
            int fee = hours * 50;
            if (hours > 5) {
                fee += 100;
            }
            System.out.println("Fee: " + fee);
        } else if (vehicleType.equals("bike")) {
            int fee = hours * 20;
```

```
            if (hours > 5) {
                fee += 50;
            }
            System.out.println("Fee: " + fee);
        }
    }

    public void showType(String vehicleType) {
        if (vehicleType.equals("car")) {
            System.out.println("Car Parking");
        } else if (vehicleType.equals("bike")) {
            System.out.println("Bike Parking");
        }
    }

    public static void main(String[] args) {
        ParkingFee pf = new ParkingFee();
        pf.calculateFee("car", 8);
        pf.showType("car");
    }
}
```

Tasks:

Identify at least 3 code smells
Explain each issue
Refactor using classes/interfaces
Improve readability